

Dynamic Pricing Models

Powered by AI

PASSENGER CAR SPARE PARTS REPRICING

Closing the weekly cost-to-price lag — and the cost-of-ownership gap it opens versus AI-enabled competitors

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PROJECT OVERVIEW

The Project Journey

This first task establishes the foundation — the full analysis follows in later stages



This deck covers the first three stages — the business challenge, the project objective and its scope — which together form the initial case idea and problem statement requested for this task.



PROJECT OVERVIEW

Project Overview at a Glance



Why This Project?

Manual, weekly repricing can't keep pace with weekly supplier cost volatility — a gap AI is well suited to close.



Business Challenge

A 3-7 day lag between supplier cost changes and price-list updates, plus informal, undocumented competitor price checks.



Objective

Explore how AI can automate repricing — from cost ingestion to guardrail-checked, published prices — closing the cost-to-price lag.



Scope

Passenger car spare parts distribution: a multi-branch warehouse distributor selling through account-tiered pricing via its ERP/DMS.



What We're Analyzing

Weekly cost-to-price dynamics, the informal competitor-intelligence process, and the resulting cost-of-ownership gap vs. AI-enabled competitors.



Expected Business Value

Lower cost of ownership, faster price reaction, reduced margin leakage and billing disputes, and a stronger “first call” position with accounts.



OVERVIEW

Pricing Methods in Spare Parts Distribution

Before the problem: how parts distributors typically set prices today

MAIN METHOD USED TODAY

Cost-Plus / Markup

A fixed % margin added over landed cost, often via a bracketed “matrix” (e.g., 55-58% GP benchmark). Simple, but static once set.

Competitive / Market-Based

Prices set to match or beat what competitors charge — requires knowing their prices, usually gathered informally.

Value-Based

Price reflects urgency or criticality to the customer (e.g., an emergency breakdown part), used selectively.

WHERE WE'RE HEADED

Dynamic / AI-Driven

Prices adjust continuously from live cost, demand and competitor data — the target state this project explores.



THE STORY

A Week in the Life of a Price List

Here's what actually happens between a supplier cost change and a new price reaching the counter:

Mon



A steel, aluminum or freight index moves overnight — landed cost on dozens of SKUs shifts

Tue– Wed



Buyers informally call around contacts at competitor branches to gauge “going rates” before reacting

Thu



A pricing clerk manually recalculates the spreadsheet, SKU by SKU, account tier by account tier

Fri



The new price list is finally keyed into the DMS and goes live — five days after the cost actually moved

By the time Friday's list goes live, Monday's cost has already moved again — the price list is stale before it's even published.



INITIAL CASE IDEA

Industry & Organization

Passenger car spare parts distribution — the automotive aftermarket

Who

A multi-branch warehouse distributor (“jobber”) supplying independent garages, workshops and retail counters with passenger car parts

Why this case

High SKU count, thin 25-40% margins, and prices that must be renegotiated constantly as supplier costs move — a textbook AI pricing problem

What's different

Unlike e-commerce, pricing here is account-tiered and manually managed inside the distributor's ERP/DMS — not yet automated



BUSINESS PROBLEM

Business Problem Statement

Supplier costs change weekly. The price list to customers is still updated manually — on the same weekly cycle, or slower.

Operational Gap

3-7 day lag between a supplier cost change and the matching price-list update

Strategic Gap

Slower, error-prone repricing weakens the “first call” position against faster, AI-enabled competitors

Financial Gap

Margin leakage while stale prices stay live, plus billing disputes from tier mismatches



THE HIDDEN COST

Cost of Ownership vs. Competitors

Manual repricing doesn't just delay prices — it carries a higher total cost of ownership (TCO) than competitors already running automated pricing.

Our manual process (higher TCO)

- Staff hours spent re-keying prices weekly across accounts
- Billing errors & credits from stale or mismatched tier prices
- Margin lost every day a price stays uncorrected
- Slower quotes — risk of losing the account to a faster rival
- Competitor prices gathered informally — phone calls, not a system

AI-enabled competitors (lower TCO)

- Automated cost ingestion — near-zero manual labor per update
- Guardrail-checked pricing — fewer disputes and write-offs
- Prices reflect current cost within hours, not days
- Faster, more consistent quotes win and retain accounts
- Competitor price signals monitored systematically, not by phone



Spotlight: The Informal Intelligence Problem

Today, the only way we know what competitors are charging is informal, friendly phone calls and personal contacts at their branches — not a system.

- Unverified & inconsistent — every buyer hears a different number, at a different moment in the cycle
- No audit trail — a price decision can't be traced back to the evidence behind it
- Single point of failure — the intelligence walks out the door when that employee leaves or the relationship cools
- Doesn't scale — impossible to keep across dozens of branches and thousands of SKUs

TASK 2

Dataset, KPIs, Before & After & Power BI Dashboard

Turning the business problem into a data-driven, AI-supported solution

Dataset → Data Cleaning → KPIs → Before vs After → Power BI Dashboard

TASK 2 FOUNDATION

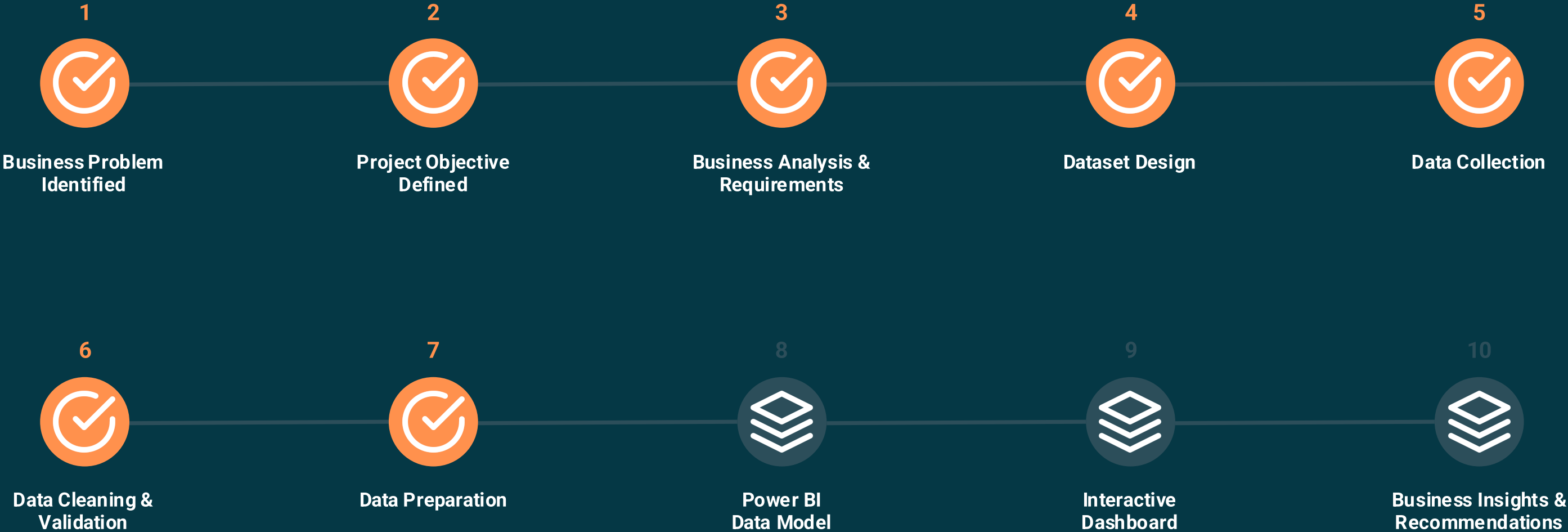
Project Progress & Data Foundation

From business problem and analysis into data preparation and Business Intelligence



PROJECT PROGRESS

Where We Are in the Project Journey



Completed



In Progress



Planned



DATASET DEVELOPMENT

From Business Problem to a Decision-Ready Dataset

The dataset was not collected at random — every field traces back to a business question this project needs to answer.



Business Problem



Business Questions



Required Data



Dataset Structure



Data Cleaning



Data Validation



Analysis-Ready Dataset



Power BI Dashboard

Why this matters

Each stage exists to answer a specific question from the business problem — the weekly cost-to-price lag and the informal competitor-intelligence gap. That traceability is what turns raw files into an analysis-ready dataset, not just a bigger spreadsheet.



DATASET OVERVIEW

The Data Categories Behind the Analysis



Product / Part Data

- Brand
- Model
- Part Number
- Part Category
- Part Description



Pricing Data

- Cost · FOB · Landed Cost
- Selling Price · Previous Price
- Competitor / Benchmark Price
- Price Gap



Sales Data

- Sales Date · Quantity
- Sales Value · Net Sales
- Channel · Customer Type



Inventory Data

- Stock Quantity ✓
- Stock Value · Movement
- Inventory Age
- Monthly Consumption



Profitability Data

- Gross Profit · GP%
- Discount %
- Margin indicators

✓ Stock Quantity loaded (23,837 parts, all 5 brands) — Days-of-Stock and movement classification are now real, computed values.



DATA DICTIONARY

What's Inside the Dataset — Field by Field

The real fields available, organized by category, with their source table

Category	Key Fields	Example	Source Table
Product / Part	Part Number, Brand, Model, Part Description, Category	“Brake Pad Front” , Hyundai	Dim_Part_All_Brands
Pricing	Current Price, FOB, Standard Cost, Landed Cost, Suggested Price, GP%	413 priced parts, 4 brands	Price_List_Master
Sales	Sales Date, Quantity Sold, Sales Value, GP, GP%, Channel, Customer Type	22,108 real sales lines	Fact_Sales_Clean
Inventory	Stock Onhand Qty, Unit Cost, Days of Stock, Movement Class	23,837 stock records	Stock_and_Cost_Master
Competitor / Benchmark	Competitor Price, Price Gap %, Segment, Generic Part Type	812 real competitor prices	Competitor_Parts_Pricing
Price History	Effective Date, List Price, Cost Change Date	497,953 real price points	Price_Change_Events

Full dataset: 21 linked tables, 600,000+ total rows across sales, pricing, inventory and competitor data.



DATASET DESIGN

Dataset Grain & Power BI Star Schema

The Grain

Not one flat table — three fact tables, each at the grain that fits its own business question:

Fact_Sales

One row = one sales order line (Part + Branch + Channel + Customer + Date)

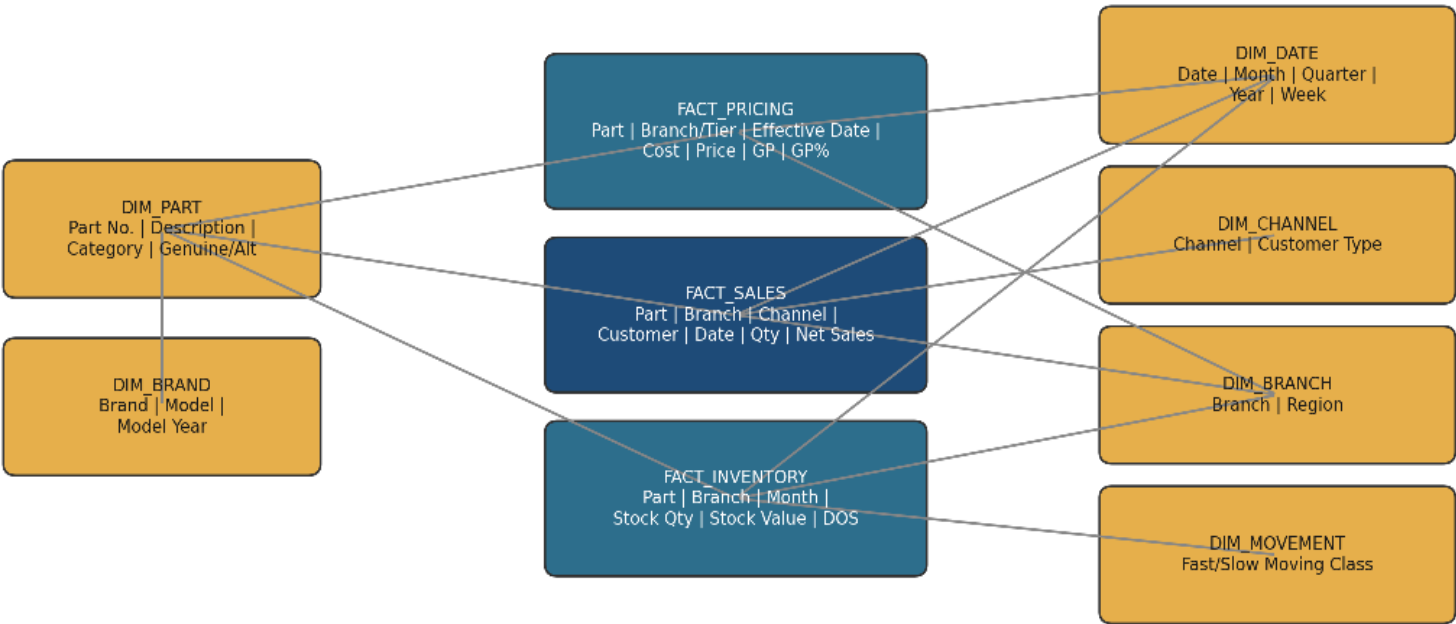
Fact_Pricing

One row = one price-change EVENT (Part + Tier + Effective Date) — needed to measure the cost-to-price lag itself

Fact_Inventory

One row = Part + Branch + month-end snapshot

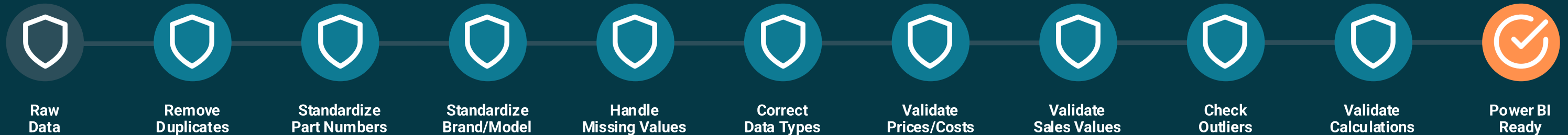
Passenger Spare Parts — Power BI Star Schema (Conceptual)





DATA CLEANING PROGRESS

Data Cleaning & Quality Preparation



Why each step matters for the business analysis

- Duplicates & inconsistent naming → without fixing these, brand/model rollups (GP% by Brand, sales trends) would silently double-count or fragment
- Missing values, wrong types & outliers → left unchecked, they distort GP%, price-gap and margin KPIs the whole project depends on
- Validated prices, costs & calculations → the Price Pass-Through Lag KPI only works if every cost and price figure feeding it is trustworthy



DATA CLEANING EVIDENCE

Before Cleaning vs. After Cleaning — What Actually Changed

Real counts from the actual source files — nothing here is estimated

Source File	Before (Raw)	Issues Found & Fixed	After (Clean)
Sales.xlsx	27,358 rows	3 footer/summary rows removed · 5,247 non-part rows (oil/labor) set aside · 684 returns flagged · 105 zero-cost rows flagged	22,108 rows (Fact_Sales_Clean)
5 Brand Master Files	Separate files, inconsistent codes	Combined into one table · Hyundai S/H prefix mismatch fixed — match rate 77% → 86%	44,485 rows (Dim_Part_All_Brands)
5 Brand Price-History Files	497,958 raw price points	5 invalid (≤ 0) prices removed	497,953 valid price events
Stock_QTY_and_Updated_Cost.xlsx	23,837 stock records	Cross-checked 2 independent cuts of the same file: 0 mismatches found	23,837 verified records

Cleaning philosophy: nothing is silently deleted. Issues are either fixed (with a documented rule) or kept and flagged for review — never hidden.



DATA QUALITY

Data Quality Framework

Every cleaning decision is checked against six standard data-quality dimensions:



Completeness

Are required fields (cost, price, sales date) populated for every record?



Accuracy

Do recalculated values (e.g., GP) match what the source data implies?



Consistency

Is the same Part Number / Brand / Model spelled identically everywhere?



Validity

Do values fall in an allowed range — valid dates, non-negative costs?



Uniqueness

Is each natural key represented exactly once where it should be?



Timeliness

Is the data current enough to support weekly repricing decisions?

Stock Quantity is now loaded — a single overall Data Quality % score will be finalized once all dimensions are formally scored. No percentage is invented here.



ANALYSIS READINESS

What This Dataset Will Enable



Pricing

- Current price vs. benchmark
- Price gaps
- Pricing opportunities



Profitability

- $GP \cdot GP\%$
- Margin opportunities



Sales

- Sales trends
- Brand / Model performance
- Channel performance



Inventory

- Slow-moving parts
- Stock exposure
- Inventory performance



Opportunity

- High-value pricing opportunities
- Margin improvement
- Priority parts/brands/models

DATASET, ANALYTICS & KPIS

Dataset, Analytics & KPIs

Dataset shape · analytics approach · KPIs & graphs to support AI-driven decision-making

Goal: preparing for simulation, starting Week 5



AI APPROACH

What Type of AI Powers This Project

Where our approach sits within the broader AI landscape — narrowing down to exactly what we use.

Narrow AI

Built to do ONE job well — not general intelligence

Machine Learning

Learns patterns from historical data, not hard-coded rules

Supervised + Unsupervised Learning

Learns from labeled price/cost history, and groups similar parts

Predictive → Prescriptive Analytics

Forecasts the right price, then recommends the action to take

Not Deep Learning, NLP or Computer Vision — our data is structured/tabular (prices, costs, quantities), which is exactly what classical Machine Learning is built for.



AI APPROACH

The Four Types of Analytics We Apply

Each type answers a different question — together they take us from data to decision.



Descriptive

“What happened?”

Historical price, cost & consumption trends from our real 2023– 2025 data



Diagnostic

“Why did it happen?”

Root-cause of margin leakage and what drives the cost-to-price lag



Predictive

“What will happen?”

XGBoost forecasts the target price; Prophet forecasts demand/consumption



Prescriptive

“What should we do?”

Optimization engine recommends the exact price to publish, within guardrails



AI APPROACH

The AI Models Powering This Project

One model per job — matched to the data we actually have, not a generic one-size-fits-all choice.

1



XGBoost / LightGBM

Supervised Learning

Predicts the target price from cost, brand, category & competitor price — trained on our real ~498K price-change events

2



K-Means Clustering

Unsupervised Learning

Groups parts by price sensitivity & movement, using real Stock & Consumption data (23,837 parts)

3



Prophet / Exp. Smoothing

Time-Series Forecasting

Forecasts monthly consumption per part from 24 months of real history — feeds Days-of-Stock

4



Constrained Optimization

Prescriptive / Decision Layer

Takes the model's prediction and picks the price to publish, within margin-floor & MAP guardrails



ANALYTICS & KPIS

The KPIs That Will Drive This Project



SIGNATURE KPI

Price Pass-Through Lag (Days)

Days between a supplier cost change and the matching price update — the number the AI pricing model is built to shrink.



Pricing & Margin

- Gross Profit (GP) · GP%
- Price Gap · Price Gap %
- Discount %
- Estimated Margin Leakage



Sales Performance

- Total Sales · Net Sales
- Sales Growth %
- Active Parts



Inventory Health

- Days of Stock (DOS)
- Inventory Turnover
- Monthly Consumption



Data & Quality

- Data Health / Quality Score
- Match Rate (Sales ↔ Master)
- Records Cleaned / Flagged

Together, these KPIs are the inputs the AI pricing engine reads to recommend and validate every repricing decision.



PROPOSED SOLUTION

AI Solution: Toward Intelligent Decision-Making

Proposed AI capabilities — not yet implemented, matched to this project's actual scope

PROPOSED

AI-Powered Pricing

- Dynamic pricing recommendations
- Price optimization
- Competitor price monitoring
- Price gap detection

PROPOSED

AI-Powered Forecasting

- Demand forecasting
- Sales forecasting
- Parts consumption prediction

PROPOSED

AI-Powered Inventory

- Slow-moving prediction
- Overstock / stock-out risk
- Inventory optimization

PROPOSED

AI-Powered Business Intelligence

- Automated anomaly detection
- Pattern identification
- Opportunity identification



THE TRANSFORMATION

Before vs. After AI

Manual → Automated · Reactive → Proactive · Historical → Predictive · Descriptive → Prescriptive

BEFORE AI

Data: Manual / fragmented

Analysis: Excel & manual review

Pricing: Reactive

Forecasting: Historical / manual

Inventory: Reactive monitoring

Insights: Delayed

Decision: Experience-driven

Simulation: Limited / difficult

AFTER AI

Data: Integrated & structured

Analysis: AI-assisted analytics

Pricing: Dynamic / optimized

Forecasting: AI-powered prediction

Inventory: Predictive monitoring

Insights: Real-time / proactive

Decision: Data + AI driven

Simulation: Scenario-based decisions



EXPECTED IMPACT

Quantifying the Expected Transformation

Directional targets only — not measured results. Nothing here is an invented percentage.

Faster Analysis

Hours → Minutes

Target

Pricing Accuracy

↑ Improved

Expected Impact

Forecast Accuracy

↑ Improved

Expected Impact

GP%

↑ Improved

Target / Scenario

Slow-Moving Inventory

↓ Reduced

Expected Impact

Decision Speed

↑ Faster

To Be Validated



POWER BI

From Clean Data & AI to an Executive Dashboard



The dashboard becomes the visual decision layer

Power BI doesn't just display the data — it surfaces what the AI models find, so management can act on it directly.



DASHBOARD DESIGN

Dashboard Visual Structure

1

Executive Overview

"What is happening?"

2

Pricing Intelligence

"Where are the pricing gaps?"

3

Sales & Profitability

"What is driving performance?"

4

Inventory Intelligence

"Where are the inventory risks?"

5

AI Insights & Opportunities

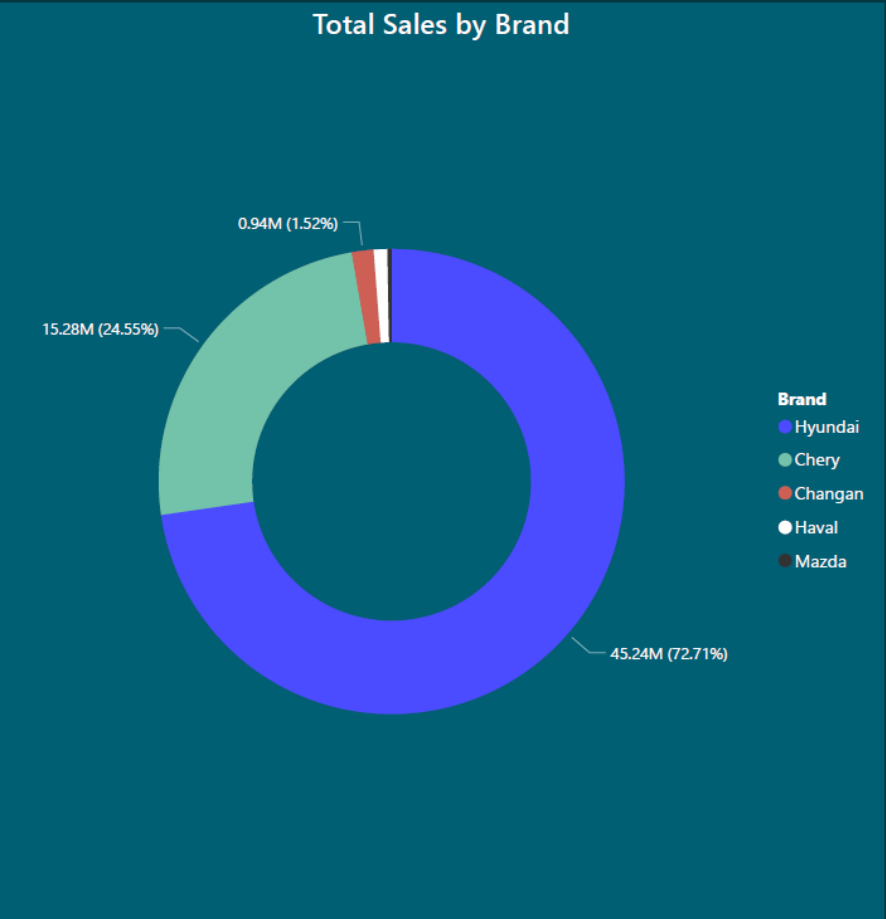
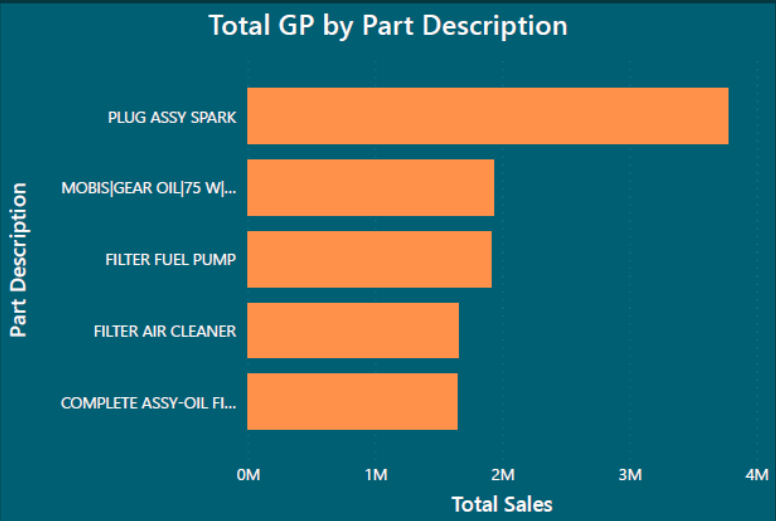
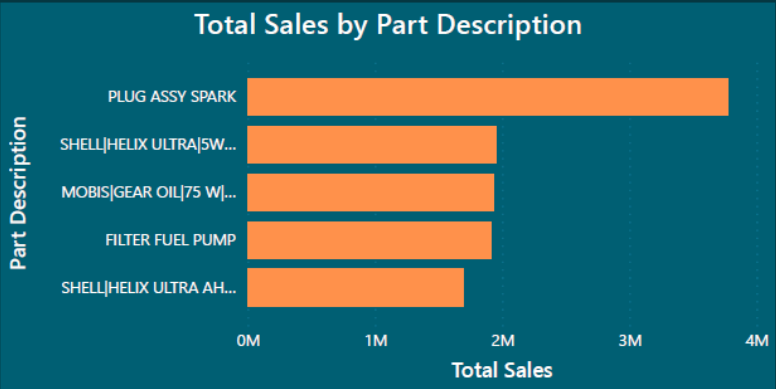
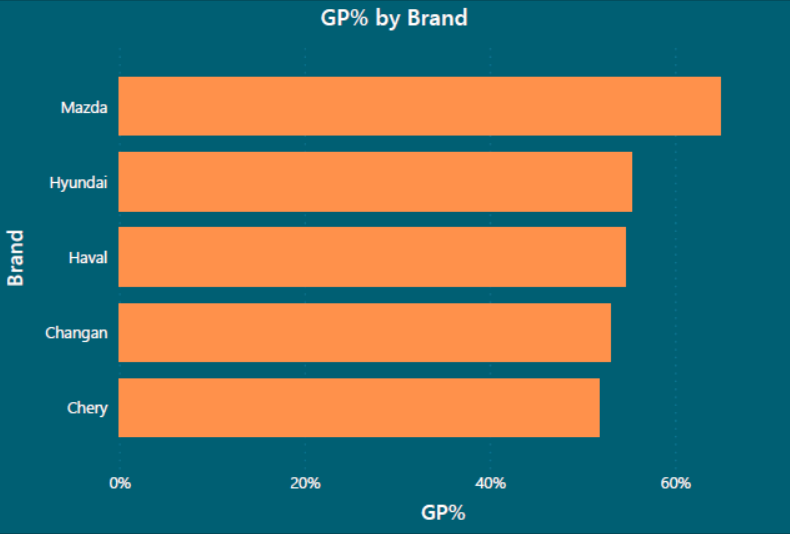
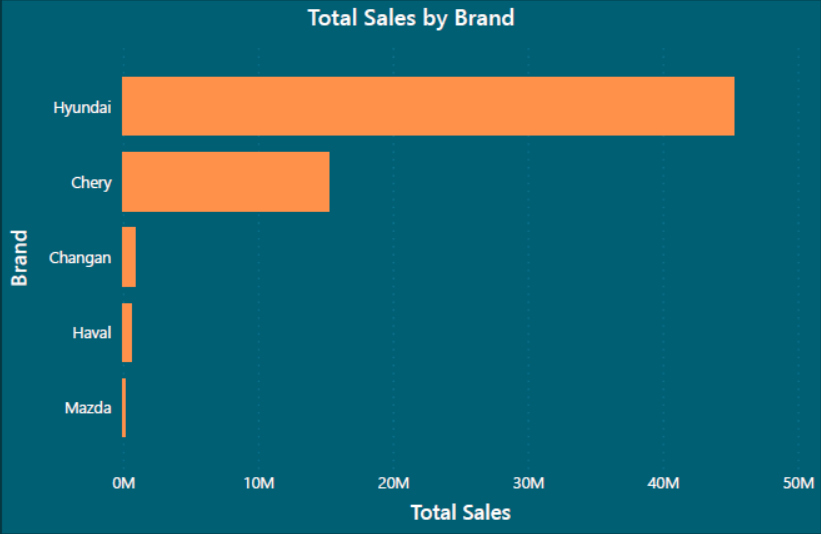
"Where should we act?"



ANALYTICS & KPIS

Dashboard Sample KPI & Graph Shapes

Dynamic Pricing Models Spare Parts



TASK 3

Simulation & Governance Framework

Moving beyond reporting into decision simulation and controlled, governed AI use

Simulation → Governance → Human + AI Decision-Making → Final Project Journey

TASK 3 DEEP DIVE

AI Transformation, Governance & Simulation

From AI-enabled insights to governed, simulated, decision-ready outcomes

Business → Data → AI → Governance → Digital Transformation → Power BI → Simulation → Decision



GOVERNANCE

AI & Data Governance Framework

How the organization controls, monitors and manages AI-driven decisions



Data Governance

- Data ownership
- Data quality
- Data standards
- Data lineage
- Data security



AI Governance

- Model monitoring
- Model accuracy
- Bias / anomaly monitoring
- Explainability
- Human oversight



Business Governance

- Pricing approval
- Exception management
- Decision authority
- KPI ownership
- Performance monitoring



Risk & Compliance Governance

- Access control
- Audit trail
- Data privacy
- Change management
- Documentation
- Segregation of duties



GOVERNANCE

Governance Operating Model

Four layers, each depending on the one below it — governance is not one checklist, it's a stack

Business Governance

Pricing approval · Exception management · Decision authority · KPI ownership

AI Governance

Model monitoring · Accuracy tracking · Bias & anomaly checks · Explainability

Data Governance

Ownership · Quality · Standards · Lineage · Security

Compliance & Control (Foundation)

Access control · Audit trail · Privacy · Change management · Documentation



GOVERNANCE

Who Owns What: RACI Accountability Matrix

R = Responsible · A = Accountable · C = Consulted · I = Informed

Governance Activity	Data Team	AI / Analytics Team	Pricing Manager	Senior Management
Data quality checks	R / A	C	I	I
Model monitoring & accuracy	C	R / A	I	I
Price recommendation review	I	R	A	C
Exception / override approval	I	C	R	A
KPI ownership & reporting	C	C	R	A
Audit trail & documentation	R	R	A	I



GOVERNANCE ROLES

Who Does What in the Pricing Governance Model



Data Team

Responsible for data quality, availability & integration



Pricing Team

Responsible for business validation & pricing recommendations



Business Management

Responsible for strategic approval & pricing policy



IT / BI Team

Responsible for Power BI infrastructure, automation & technical support



AI / Analytics Team

Responsible for model performance, monitoring & improvement



CONTINUOUS GOVERNANCE

The Governance Control Loop — Not a One-Time Project



Data
Collection



Data
Validation



Analytics



AI
Recommendation



Business
Validation



Approval



Implementation



Monitoring



Performance
Review



Model / Rule
Improvement

↻ Feeds back into Data Collection — a continuous cycle, not a one-time project



HUMAN + AI

AI Enhances the Decision — It Does Not Replace It



Data



AI
Recommendation



Business
Validation



Management
Approval



Business
Action



Performance
Monitoring

The human stays in the loop

Every AI price recommendation passes through business validation and management approval before it becomes a live action — AI accelerates the decision, it doesn't bypass governance.

↻ Performance results feed back into the AI / Analytics team — closing the loop for continuous model improvement.



DIGITAL TRANSFORMATION

Five Types of Transformation — Applied to Our Pricing Process

People + Process + Data + Technology + Decision-Making, working together to close the pricing gap

Process

CURRENT
Manual pricing review, fragmented analysis

FUTURE
Standardized, data-driven pricing workflow

Data

CURRENT
Fragmented, inconsistent sources

FUTURE
Clean, structured, integrated pricing dataset

Technology

CURRENT
Manual tools, disconnected reporting

FUTURE
Power BI + analytics + AI-enabled decision support

Decision-Making

CURRENT
Reactive, experience-based

FUTURE
Data-driven, predictive, scenario-based

People & Organization

CURRENT
Team spends most time collecting/preparing data

FUTURE
Team focuses on analysis, validation & strategy



DIGITAL TRANSFORMATION

From AI Solution to Digital Transformation

Data → Analytics → AI → Automation → Digital Decision-Making → Continuous Improvement

Level 1: Manual

Level 2: Digital Data

Level 3: Business Intelligence

YOU ARE HERE

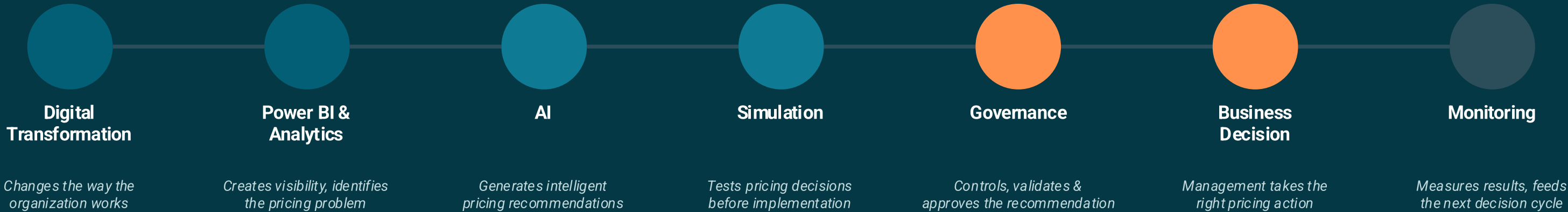
Level 4: AI-Powered Analytics

Level 5: Predictive & Prescriptive Decisions

TARGET

THE CONNECTION

How Transformation, AI, Simulation & Governance Work Together



Digital Transformation enables the new way of working. AI provides intelligence. Simulation evaluates alternatives. Governance ensures control. Management makes the final decision.



SIMULATION

Business Simulation & Analysis

Test scenarios before making decisions — variables management can adjust

Pricing

- Price increase %
- Price decrease %
- Target GP%

Demand

- Demand growth %
- Demand decline %

Cost

- Cost increase %
- Cost reduction %

Discount

- Discount %

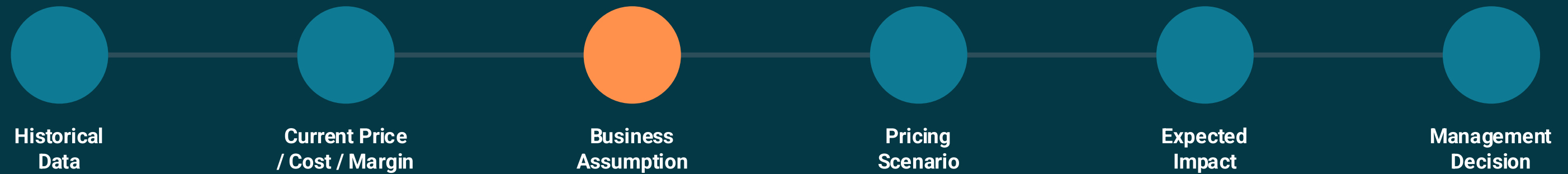
Inventory

- Demand change
- Stock level
- Reorder parameters

Calculates impact on:

Sales · Gross Profit · GP% · Quantity · Inventory Value · Margin · Business Opportunity

SIMULATION METHOD Pricing Simulation



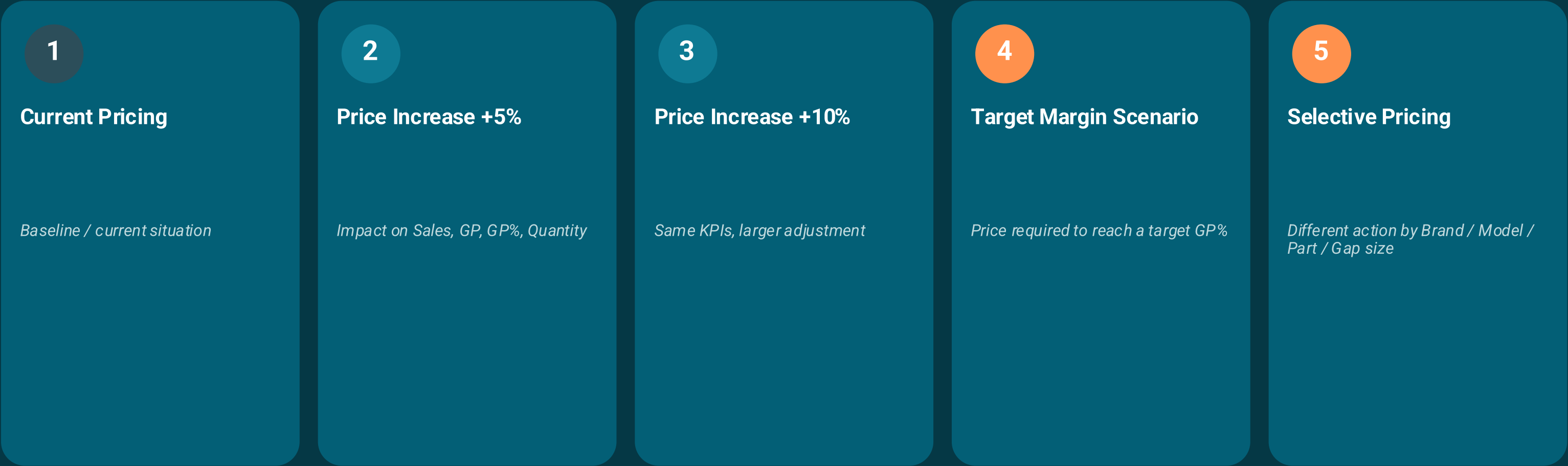
Why the “Business Assumption” step matters

This dataset does not contain a measured demand-elasticity figure (how quantity responds to a price change). Every scenario in this section discloses its assumption explicitly – nothing is presented as a measured fact. Demand response should be validated using historical elasticity data before implementation.



SIMULATION SCENARIOS

Five Pricing Scenarios — Not One-Size-Fits-All



“Right Price → Right Part → Right Scenario” — we should not apply one uniform price increase across all parts



FULL SIMULATION (ANYLOGIC)

Built in AnyLogic: two correction paths simulated over time, from the same real starting problem

A Real System Dynamics Simulation

-80.8%

STARTING GAP (Real)

18 days

TRADITIONAL LAG (Real)

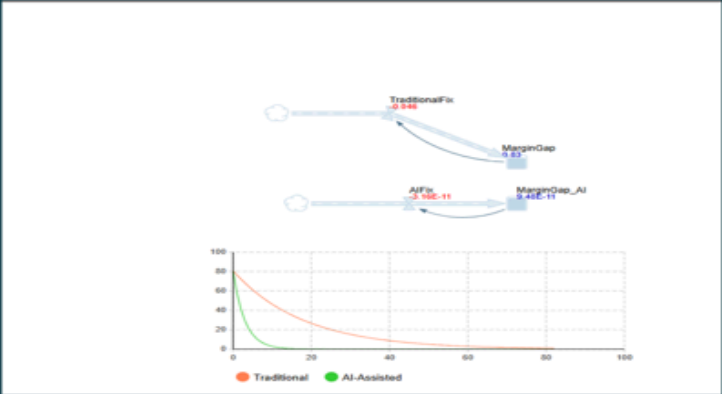
3 days

AI-ASSISTED (Assumption)

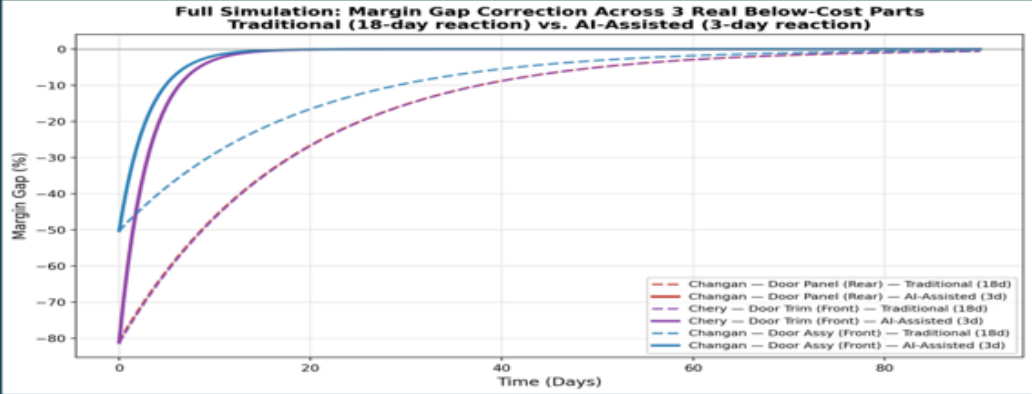
94.9%

P(AI FASTER) (Monte Carlo)

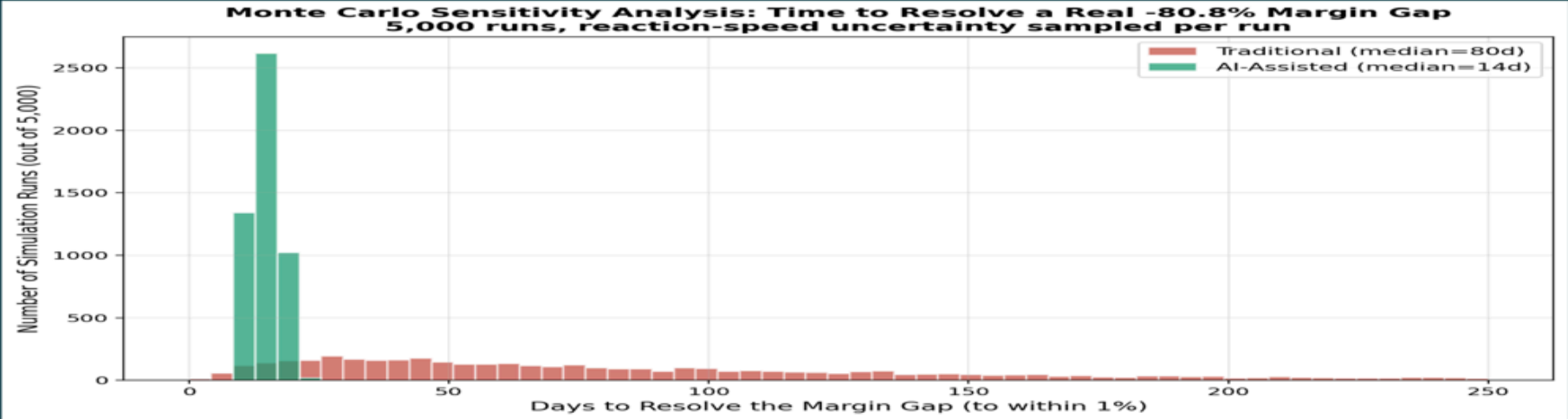
A. Built in AnyLogic — System Dynamics Model



B. Same Model Applied to 3 Real Below-Cost Parts



C. Monte Carlo Sensitivity Analysis — 5,000 Simulated Runs



What This Simulation Shows

- Starting point: a real margin gap of 80.8% (the Changan door panel case)
- Traditional correction (orange): improves gradually, following the measured 18-day reaction pattern — still not resolved after 70+ simulated days
- AI-assisted correction (green): resolves to near-zero in roughly 15– 20 simulated days
- Built as a System Dynamics model (Stock & Flow), the result unfolds over simulated time



SIMULATION DEEP DIVE

Inside the AnyLogic Simulation — What We Actually Built

Model Type	System Dynamics (Stock & Flow) — chosen because the goal was to show a problem closing over time, not a single calculation
The Stock	“MarginGap,” initialized at – 80.8% — the real, measured margin on the Changan door-panel case
Flow 1: Traditional Fix	Correction rate = – MarginGap / 18 — modeled directly on our measured 18-day median price lag
Flow 2: AI-Assisted Fix	Correction rate = – MarginGap / 3 — the target reaction speed under an AI-recommended workflow
Output	A time plot comparing both paths from the same starting point — AI-assisted reaches near-zero in ~15– 20 simulated days vs. 70+ for the traditional path

Scope note: the model is initialized from one real, representative case (the Changan door panel), not a bulk import of the full dataset — stated here for academic transparency.



SIMULATION MATH

Simulation Method & Assumptions

Current Price × (1 + Scenario Price Adjustment %) = Simulated Price

Then recalculated: Revenue → GP → GP% → Quantity → Margin





SIMULATION

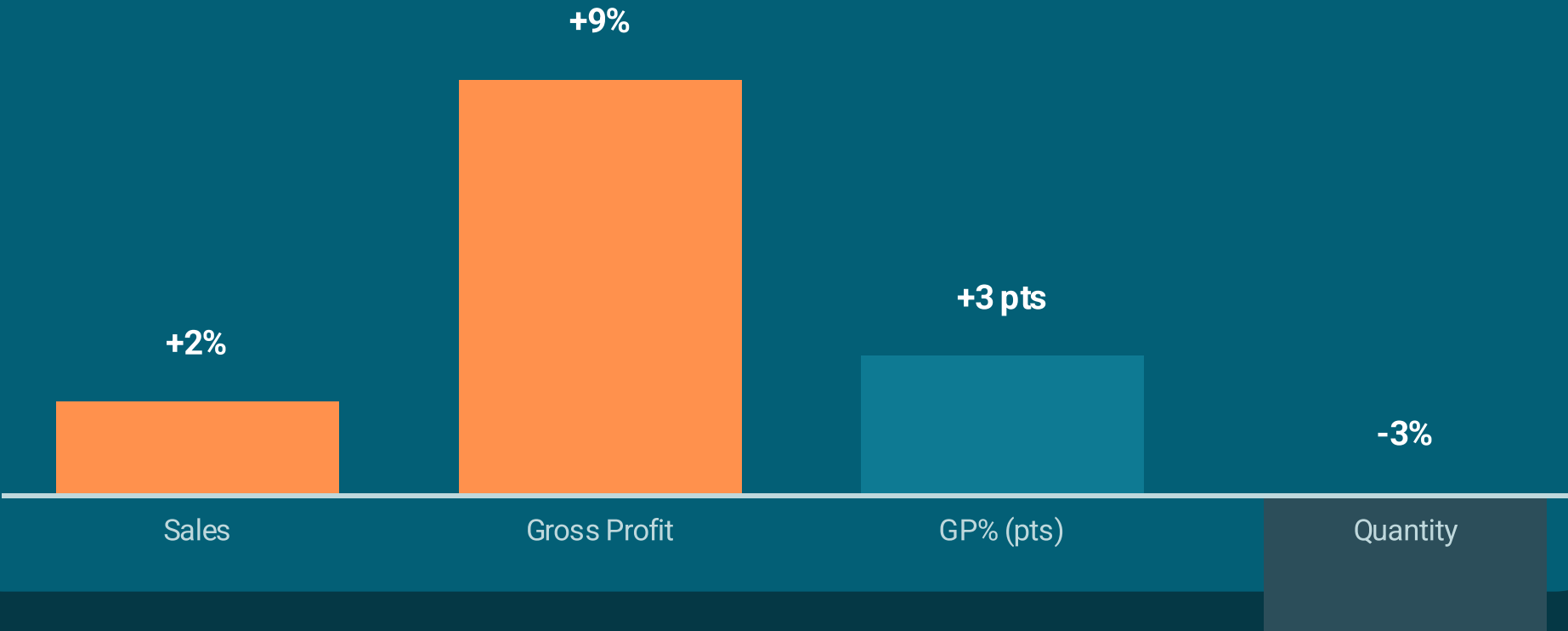
Simulation Example: A Price Scenario Walkthrough

ILLUSTRATIVE SCENARIO — NOT ACTUAL DATA OR RESULTS

Scenario Inputs (user-adjustable)

Price change	+5%
Target GP%	45%
Demand sensitivity	Medium
Cost change	0%
Applies to	Oil Filters — Chery

Illustrative Impact (What-If, not measured)



This is what the simulation page in Power BI will let management explore interactively — dragging sliders like these, before any real price change is approved.



SIMULATION OUTPUT

Scenario Comparison — Authorized Centers & Outlets

CURRENT = ACTUAL DATA. ALL SCENARIO COLUMNS = SIMULATED, USING A DISCLOSED ASSUMPTION

KPI	Current (Actual)	Scenario 1 +5% Price	Scenario 2 +10% Price	Scenario 3 Target Margin 60%
Sales (M EGP)	62.2	64.3	66.4	62.2 (price only)
GP (M EGP)	33.8	36.4	38.9	37.3
GP%	54.5%	56.6%	58.6%	60.0%
Quantity	54,573	53,754	52,936	54,573 (unchanged)

Assumption disclosed: mild demand sensitivity of – 0.3% quantity per +1% price (not measured in this dataset)

A real example: uniform increases don't fix every part

A Changan door panel is currently sold at 10,970 EGP against a cost of 19,837 EGP (– 80.8% GP%). Even a +10% increase (12,067 EGP) leaves it deeply negative. Reaching breakeven requires pricing at cost (19,837 EGP) — an – 80%+ correction on this specific part, not a company-wide rule.



SIMULATION OUTPUT

Before vs. After Simulation



Current Price



AI Recommended Price



Expected Sales Impact



Expected GP Impact



Expected GP% Impact

Every downstream box is a simulated projection, clearly separated from the historical, measured data everywhere else in this presentation.



BUSINESS VALUE

What This Solution Can Deliver — Connected to the Business Problem

Pricing accuracy

Fewer parts priced below cost or below target margin

Margin protection

Below-cost items (like the 15 identified) caught and corrected

Revenue opportunity

Top pricing gaps quantified (e.g., ~16.1K EGP top single-part opportunity)

Decision speed

Faster reaction than today's 18-day measured price lag

Management visibility

Real-time Power BI view vs. periodic manual reporting

Scenario planning

Test price changes before committing to them

Language used throughout: “Potential impact” and “Expected improvement” — not guaranteed results until validated by implementation.



THE TRANSFORMATION

Where We Were, and Where We Are Now

BEFORE AI
Price reaction time: 18 days (median, measured)
Pricing gap visibility: Manual review, hit-or-miss
Below-cost parts: 15 parts undetected for weeks
Decision method: Experience-based, reactive
Testing before action: Not possible
Governance: Informal, undocumented

AFTER AI
Price reaction time: 3 days (target, simulated)
Pricing gap visibility: Automated dashboard, 100% of parts
Below-cost parts: Flagged immediately on data refresh
Decision method: AI-recommended, simulation-tested
Testing before action: What-If + full AnyLogic simulation
Governance: RACI-defined, auditable, continuous



BUSINESS CASE

Return on Investment: Cost, Productivity & Accuracy

INPUTS MARKED (REAL) ARE MEASURED; INPUTS MARKED (ASSUMPTION) ARE DISCLOSED ESTIMATES, NOT MEASURED FACTS

Cost Reduction	Productivity	Accuracy
- Pricing team: 5 staff × 15,000 EGP/month (Real, given) = 900,000 EGP/year - Assumption: ~40% of team time spent on the manual weekly repricing cycle - Assumption: AI/automation removes ~65% of that manual effort - Estimated capacity freed: ~1.3 FTE-equivalent ≈ 234,000 EGP/year — redeployable to higher-value pricing analysis, not necessarily headcount cuts	- Price reaction time (Real, measured): 18-day median lag today - Target with AI-assisted workflow (used in simulation): 3 days - Estimated reaction-speed improvement: ~83% faster - Basis: same correction-rate logic validated in the AnyLogic simulation (Slide “Inside the AnyLogic Simulation”)	0 discrepancies (Real, measured) found recalculating GP independently across all 22,108 real transactions 15 below-cost parts (Real) had gone undetected for weeks under manual review - With automated flagging, this class of error is caught on the next data refresh, not weeks later



THE JOURNEY

The Problem — and How AI Solved It, Step by Step

THE PROBLEM: Supplier costs move weekly; prices update on a manual, ~18-day median lag — causing margin leakage, including parts sold below cost.

1

Clean the Data

27,358 raw sales rows → 22,108 validated rows; every fix documented, nothing silently deleted

2

Build the Model

Star-schema dataset + Power BI dashboard — pricing gaps and below-cost parts became visible for the first time

3

Apply AI Models

XGBoost price prediction, K-Means segmentation, Prophet demand forecasting, constrained optimization

4

Simulate Before Acting

What-If scenarios and a full AnyLogic System Dynamics model tested the fix before any real price changed

5

Govern the Decision

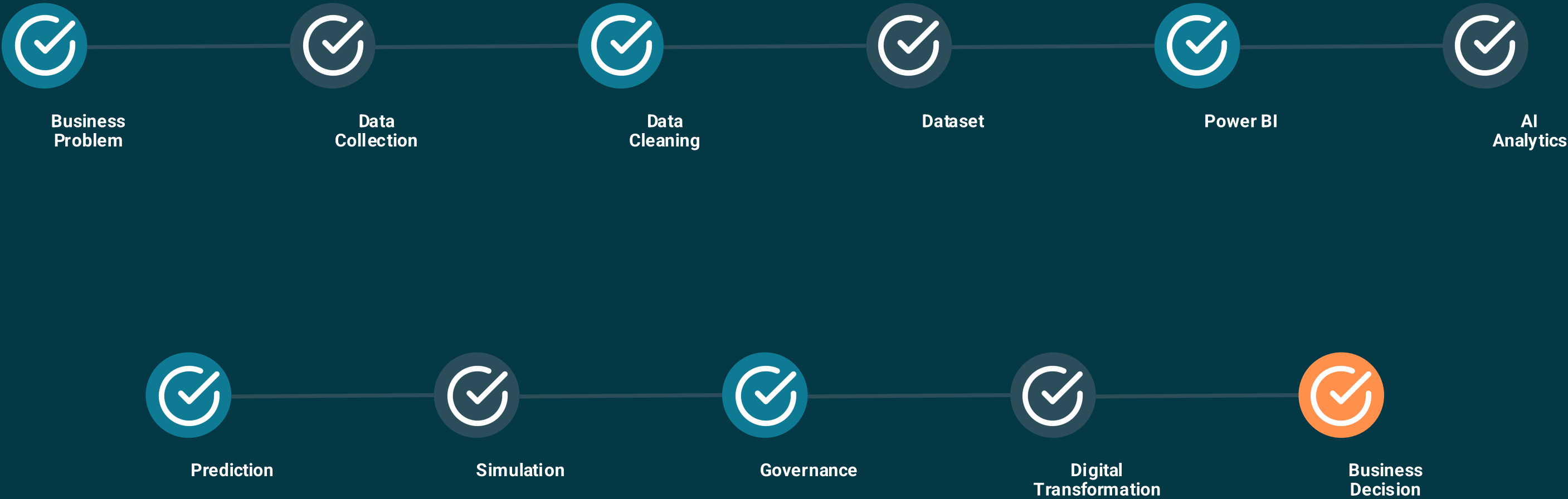
AI recommends; the pricing team validates; management approves — human control preserved throughout

THE RESULT: A real, working system — built, tested, and documented end to end, not just proposed



THE FULL JOURNEY

From Business Problem to AI-Powered Decisions



Transforming Passenger Spare Parts Management from Reactive Reporting to AI-Powered, Governed, Data-Driven Decision Making.

Thank You

Questions & Discussion